

Math 526, S12

Quiz 2

Section:

Name: *Answer*

SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (4 points) Compute the multiplication of the two matrices

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 0 \\ 1 & 3 & 1 \end{bmatrix} \\ = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

Problem 2. (6 points) Write down the  $3 \times 3$  elimination matrices that produce the following operations:

(a)  $E_{21}$  subtracts 5 times row 1 from row 2.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)  $E_{32}$  adds 7 times row 2 to row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$$